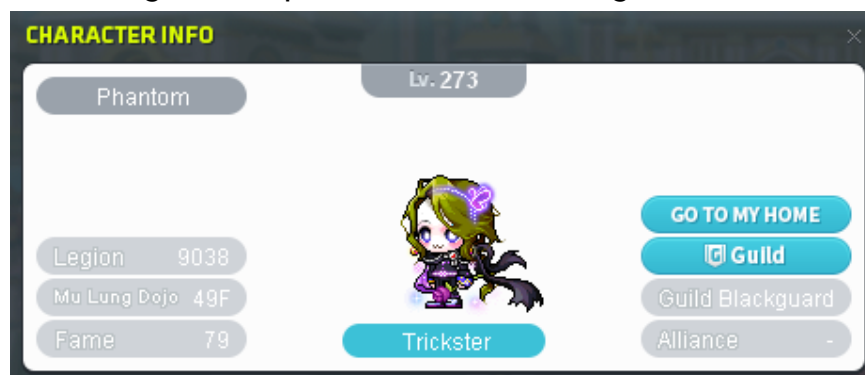


Maplestory's Big 3

As a preface, I will not really be talking about my characters or related progress updates - instead this blog will focus on the 3 big game progression mechanics that have been overhauled since 2020. Game knowledge isn't required but basic probability/cs knowledge will help a lot in the Starforcing section.



My Reboot (Kronos) Main

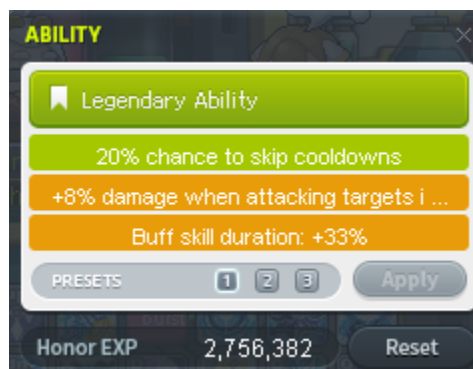
Let's start with a really high level overview of the game: A Korean 2D MMORPG that nowadays has embraced two gameplay loops. The first is defeating bosses to acquire desired loot, selling or using the loot to gain wealth, and spending the wealth to defeat tougher bosses. The second is defeating monsters around the world to gain EXP and rare loot that only monsters (not bosses) drop; from there you can either unlock new areas/bosses/monsters or sell the loot to speed up your bossing gameplay loop. If you've read any powerscaling manhwas then this might sound familiar and in fact I'm sure your favorite manhwa author has played Maplestory at some point because in South Korea it's up there with League of Legends in popularity.



So what avenues of powerscaling does the game provide? While there's a million systems (no really there's honestly way too many) we're going to focus on 3 in particular: Inner Abilities (IA), Item Potentials (Pots/Cubing), and Item Starforcing (SF).

Inner Abilities

Basically your character has 3 unique stat lines that can be rerolled with a custom currency called honor. While you can reroll all 3 lines like a slot machine, most players end up "locking" 1 or 2 slots which allow you to only gamble for the last line independently. The goal of most players is to obviously pick the best 3 lines for their particular character which ends up being 3 lines in the Legendary (Green), Unique (Yellow), Unique value brackets as well as the actual stats desired.



Phantom Class IA

For most characters this mechanic just provides a small stat buff but specific character classes end up being defined by their IA and it would be really awful if the IA reroll function had some crippling programming error...

So in 2020 a few Korean players had suspicions about the reroll function and started gathering data based from various streamers. Eventually it was determined that you could abuse the IA reroll function via something called the "Sandwich Method" and by extension reverse engineer the entire reroll system. The sandwich method was an observation that locking the first and third lines of your IA produced an uneven distribution of results in the middle line. Of course this works both ways where some players could never hit their final optimal IA line while others couldn't whiff even if they tried.

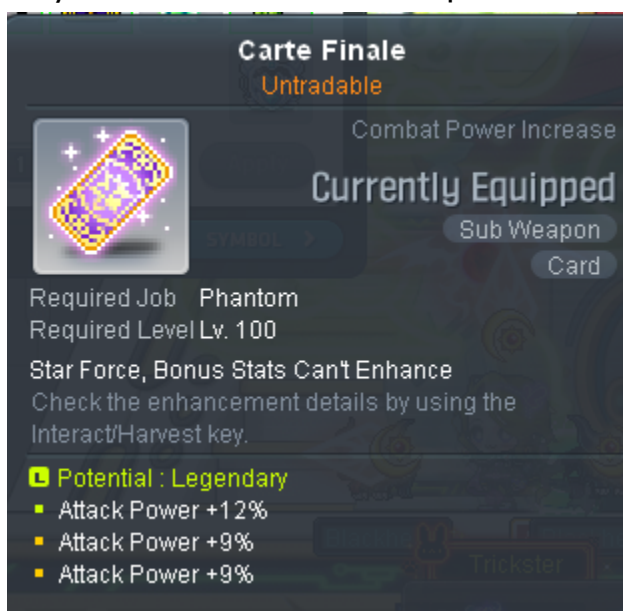
After this method went viral Nexon put out a few statements talking about the IA reroll system (under the looming threat of the Korean FTC) as well as future plans to rework the reroll function. If you're curious here are some translated posts: [Link 1](#), [Link 2](#).

While this system isn't incredibly powerful in the grand scheme of things, it's just a small peek into the 25 year spaghetti storm that plagues Nexon developers every day.

Before we continue onto the real meat and potatoes of this blog I want to give an honorable mention to flames since I completely forgot those were ALSO rigged to favor STR characters... From memory it's because flames used to pick two from the top and two from the bottom of a string list which heavily favored STR and All Stat lines together versus any of the other 3 stats. It's why a lot of legacy flames had Main Stat + All Stat + Jump + Speed...

Item Potentials

While a lot less interesting from a programming perspective, cubing (the act of rerolling item potentials) is one of THE central ways a character can powerscale. In the Global servers (GMS) you need to use special items to reroll 3 lines similarly to IA. The main difference is that there's no locking - you have to just hope you hit the jackpot. While there are weird cubes that alleviate the variance at the end of the day you will be clicking "Try Again" until you hit that sweet sweet 3L pot.



3L Attack which will get transferred to Astra secondary ASAP

Like with IA all the spreadsheet warriors have calculated the optimal potentials for the important items (Weapon, Secondary Weapon, Emblem AKA WSE) and in GMS there really isn't a need for any WSE pot to have triple Boss Damage (eg. 40% BD, 30% BD, 30% BD). However after the IA scandal there was a brigade of dedicated data gatherers in Korea and eventually it was discovered (through the official data API) that no player has ever rolled triple boss. Ever. Out of the trillions and trillions of WSE pots produced. So once again the Korean FTC came knocking and it turns out that triple

boss was hardcoded to never appear. Why? The hater in me suspects that the “game designers” *thought* triple boss would be too strong but never actually verified numerically. Anyways, this one actually got the director at the time (Wonki) booted and he was placed in mobile game jail (where he proceeded to do the same scandal in Maplestory Idle which then got him obliterated).

Before we go onto the last (and actually non-scandalous) mechanic I do want to pour one out for the developers because as someone who has gone through the trenches of audit it’s shocking just how many things can slip through the cracks. While it’s easy for a player to assume all these scandals were done out of greed or malice I think it’s easier to believe that no one wanted to go through decades old code to make sure it was coded “properly” - after all they only have so much time to deliver their actual features and wasting any time on reviewing code that seem to work must feel silly. Of course I’m not abdicating blame from the original developers since they definitely have no excuse but in the eyes of audit the only thing that matters is identifying and assigning issues...

Item Starforce (Starforcing)

Ahh starforcing, a mechanic so annoying it made me stop playing for a hot minute. If you’ve played Black Desert Online or Lost Ark then this might sound familiar - an independent item enhancing mechanic where you spend resources for the chance of gaining more stats. Maplestory’s twist (back in 2020) was that you had 3 major breakpoints: Out of the 25 stars the mechanic changes at 11 stars, 18 stars, and 23 stars (it’s the 17 stars and 22 stars).



Just wanna flex this item...

Going from 11 to 15 every failure kicks you back an entire star (with the floor being 10 stars). Then once you hit 15 your star floor becomes 15 stars but once you hit 18 stars your items can randomly explode. Yep. Explode. While technically they can explode from the 13th star and up, the “safeguarding” feature lets you remove that chance for

an additional fee per attempt. Assuming your item doesn't explode and you hit 20 stars then you've hit the last star floor and there's just two consecutive successes between that item and being considered endgame. If you're wondering about the 23rd to 25th star then that's basically for insane whales since the success is 3%, 2%, and 1% with basically guaranteed odds of item destruction on failure...

Fun trivia fact, BTS Jin rage quit Maplestory after he blew up a rare item on the first attempt on the 13th star! (It was a Dominator Pendant back when that was BiS).

Also in case any reader DOES play the game I do know about the 30 star update and Inkwell's janky protection thing but honestly the concepts are the same...

Ok going back on track, starforcing is the ultimate powerscaling mechanic and as a result there's a lot of importance in understanding the risks, costs, and odds for your character. While heuristic starforce calculators like Suckhard/BrandonMay do the job, some folks wanted an analytical solution so instead of studying for uni exams I instead coded a Python script. If you're not interested then feel free to scroll to the bottom of the blog since there's a lot of #math in this section...

Starforcing Calculator

If I was going to make an analytical solution I needed to find an abstract model which fits the starforcing mechanic. While there's likely a lot of different models I picked a (not-so) Hidden Markov Model for 2 main reasons: I was used to the concepts after taking a few uni classes on machine learning and I knew its base properties matched this game mechanic perfectly:

- We only care about an item's current state (t) and subsequent state ($t+1$) eg. whether it booms, succeeds, fails, or fails (with star demotion)
- Starforcing is inherently discrete since you cannot have fractional stars
- Since most players care about the probability of hitting a desired state we can implement a check to cut off the computations early which saves time on the iterative algorithm

So with the abstract model and desired application in mind I implemented the following algorithm based on the HMM forward-propagation dynamic programming concept:

1. Ingest Starting Star (s_0), Target Star (s_1), desired probability of success (p) as well as optional parameters (star catching, SSF, safeguarding, etc)
2. Load cost table, probability table, and HMM where $HMM[s_0] = 1$
3. Begin the iteration process by first checking if $HMM[s_1] \geq p$, if not then go in.
 - a. Apply the probability table on the current HMM model starting from s_0 up to s_1 with the results stored in an intermediary table. The final value at

star $s_0 + 1$ would be: $\text{temp}[s_0 + 1] = \text{HMM}[s_0] * p(\text{success})[s_0] + \text{HMM}[s_0 + 1] * p(\text{fail-stay}) + \text{HMM}[s_0 + 2] * p(\text{fail-drop})$. Of course since we're iterating by star level this final value is computed over 3 steps.

- b. Compute the expected cost of a "tap" (attempt) by storing the dot product of our current HMM and the cost table. This value is appended to a final global value that stores the expected cost.
 - c. Compute the expected "boom" (item explosion) per tap by accumulating all $p(\text{boom})$ values calculated earlier in the function.
 - d. Reassign the newly computed temp table to HMM (or if you want to save memory you can compute each step within the same array). Since we know the probability of s_1 monotonically increases we can immediately recheck $\text{HMM}[s_1]$ to see if we've reached the desired probability (eg. 0.25, 0.5, 0.75)
4. Print out the results in a pretty way including quartile statistics

Some notes I'll mention in case the reader plays the game (or is a nerd)

- In GMS Lightning you don't have to worry about $p(\text{fail-drop})$ and you could just add some additional parameters (or hardcode) for Inkwell's weird boom protection slider. I haven't rewritten my script but from what I've read it's pretty bad unless you're working on pitched+ gear
- Booms are implemented as increasing the probability of being at star 12. I think there's supposed to be a patch to change that but once again you'd just add this as a parameter or hardcode the change based on heuristic assumptions
- We know the model will eventually converge because we don't ever touch the final star s_1 so every successful 30% tap from $s_1 - 1$ to s_1 will be terminal. If you want an intuitive example think of a one-way chemical reaction and how the final product doesn't influence the current rate of change
- For the actual source code I put it on pastebin years ago:
<https://pastebin.com/Run6wjDE>

Summary

While I personally still think OSRS is the best MMORPG ever created, I will give Maplestory credit since the current director Changseop has done a LOT of work resolving pain points and restoring Nexon's reputation after the consecutive RNG scandals. Quality-of-Life through the wahzoo, Challenger Worlds akin to leagues, reworking antiquated mechanics, and properly balancing classes have all been well received. But the ball keeps rolling as some experimental features are rolling which could push the game further beyond (I'm coping SO hard for iPad support but sadly most of the crazy features are only for Korean servers). I'd personally contribute most

of the recent successes to the game developers who clearly understand what game loop Maplestory is, what it should be, and where it should go.

As for the GMS director Inkwell... he's trying his best with RoD and Althea... I guess I'm mostly just glad they didn't shutdown Reboot (Kronos nerd emoji)

And if you're curious on what my plan is for this major summer patch, I'm putting all my meso into mechaberries so if I don't hit 280 before they expire I might just hamter implode...



